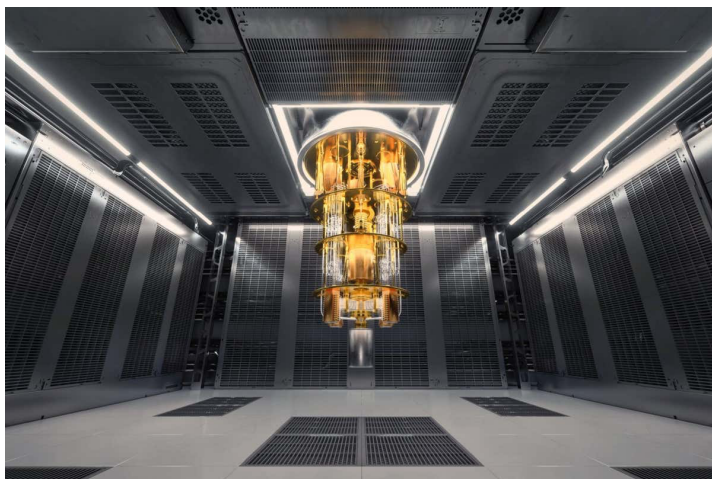


To what extent will quantum computation develop into a transformative technology within the next decade?

Daniel Fussey, EPQ Dissertation 2024



(Wilkins, 2023)

Abstract

Quantum computing has been the source of great media hype and investment over the past decade, despite having provided few apparent returns. This paper aims to examine whether, in the next 10 years, quantum computing will have a transformative technological impact, or whether the subject will remain the technology of the “near future”, forever promised yet never delivered. This paper discusses both the physical roots of quantum computation’s benefits, as well as providing an overview and analysis of current quantum computational applications. By analysing both areas, the author aims to predict the viability of quantum computation, and thus gauge its potential impacts.

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1 Introduction

Quantum computers were first proposed by Richard Feynman and Paul Benioff in 1981 (Feynman, 1986), designed to exploit the intrinsic quantum properties of quantum physics in order to achieve computational gains potentially orders of magnitudes greater than their classical counterparts. Though initially conceived for scientific purposes – simulating the evolution of a quantum system (a task impossible for a classical computer to accurately perform), quantum computers have an unrealised potential in a large variety of other fields. Many large corporations and well-funded start-ups have emerged in order to try to exploit this, from Google and Microsoft to IBM (Gil', et al., 2023). Whilst recently, quantum computing has often been portrayed within the media as the next big leap in computing, potentially ushering in a new technological era just as their classical counterparts did in the 1990's, these claims have been consistently voiced across the past decade, with seemingly little actual progress (Gil', et al., 2023).

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This paper aims to understand the transformative ability of quantum computing as both a viable alternative to classical computation, as well as the potential for quantum computing to open up new possibilities beyond the realm of classical computers. In order to be classed as a viable alternative in any one field, the author seeks to determine whether quantum computers have any *practical* benefit over pre-existing classical computers, whether this benefit is outweighed by other mitigating factors such as running costs, and whether the technology is likely to be successfully developed and implemented within the next decade. However, determining the potential for quantum computation to conduct novel tasks is considerably more challenging – just as few early developers of the computer might have predicted the online revolution of the computer-enabled internet, current proponents of quantum computing cannot hope to fully gauge its impact. Whilst the future is inherently difficult to accurately determine, by first analysing the fundamental building blocks of quantum computers, this paper aims to understand where the theoretical difficulties will lie within any future quantum computer, given the necessary assumption that the most fundamental operating principles of quantum computer construction shall remain unchanged. Furthermore, by analysing the fundamentals of quantum computation, it is also possible to understand where and why quantum computers might have any chance of superseding classical computation, known as quantum supremacy. However, this paper does not discuss quantum supremacy itself, which is a highly theoretical benchmark between the most powerful classical supercomputers and the most powerful quantum computers, but rather looks to discuss possible *practical* supremacy, which though harder to quantitatively gauge, is the key determining factor in the adoption of quantum computers.

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The discussion may be decomposed into two key sections:

1. The fundamental mechanics behind quantum computer operations, and an analysis of the resulting benefits and limitations.

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2. The possible viable applications of quantum computing, and how they compare taking the classical computers used for each task as qualitative benchmarks.

Each section discusses both the potential for improvements resulting from quantum technology, as well as its limitations.

It is only by appreciating both where any quantum gains emerge from, and what the quantum benefits can conceivably be used on, can we hope to discern the future of quantum computation.

2 Literature Review

2.1 Quantum computing: a taxonomy of systematic. Review and future directions.

Gill, SS, Kumar, A, Singh, H, et al. Quantum computing: A taxonomy, systematic review and future directions. *Softw: Pract Exper.* 2022; 52(1): 66–114. doi:[10.1002/spe.3039](https://doi.org/10.1002/spe.3039)

The authors in this peer reviewed paper discuss the basics of how quantum computers operate, systematically evaluating each component in areas such as feasibility, difficulty of implementation and technical viability. It details the main building blocks required to achieve a successful large-scale quantum processor, like quantum gates, error correction and quantum memories, giving concise discussions of the different approaches researchers are taking on constructing each component. Furthermore, the authors discuss possible applications of quantum computers, such as improved computational models for chemistry, and weather, detailing how quantum computing could more efficiently perform tasks that classical computers struggle to solve exactly. The paper then goes on to analyse the bigger picture of the current quantum computing landscape, providing readers with an overview of the cutting-edge quantum technologies that are enabling the quantum advantages. The paper is most useful as a source of quantum computing enabled technologies such as its potential applications to machine learning, for, as authors have analysed and categorised these technologies, it facilitates direct and insightful comparisons between multiple otherwise differing applications of quantum computers. Moreover, the paper provides analysis of the stages of development that these technologies are at, for example, providing a hype cycle diagram for quantum computing.

However, the authors do little to analyse the potential impacts on the world of the quantum enabled technologies, rather simply listing them, and providing explanation of how they work. This practice of concise explanation with little impact analysis is symptomatic of the entire paper's approach to the topic, suggesting it is primarily aimed at academics looking to enter the quantum computing field or academics looking for a review of current technologies, rather than an analysis of their impact. Moreover, due to the brevity of each section, and the wide scope of the study, this paper acts only as a summary of the current cutting edge, so has limited use in understanding how certain quantum technologies would work. This means that the paper serves more use as a springboard for future research directions, providing an overview of potentially crucial quantum technologies, as well as giving warning to over-hyped technologies which are unlikely to emerge within a 10-year timeframe.

The paper comes from a trusted peer-reviewed journal (Journal of Software: Practice and Experience), and, as of December the 5th 2023, has 55 citations, having been published in January of 2022. This shows that it has rapidly proven useful in the quantum field, suggesting at a reliability of methodology. Furthermore, the recent publication date ensures that any information given in the paper should not be out-dated. This is crucial given how quantum computing is a fast-moving field, so any useful systematic taxonomy of the subject must be within the last 2 years. Due to the journal's peer review status, it can be assumed that the data and information given is valid, reinforcing trust within the paper.

2.2 An Introduction to Quantum Computing for Non-Physicists

*Eleanor Rieffel and Wolfgang Polak. 2000. An introduction to quantum computing for non-physicists. ACM Comput. Surv. 32, 3 (Sept. 2000), 300–335.
<https://doi.org/10.1145/367701.367709>*

This paper has the stated aim of providing an introduction into quantum computing for non-physicists. Though some level of technical background is assumed, with a required proficiency in linear algebra to progress through certain sections, overall, the paper achieves its aim. The paper initially discusses qubits (quantum equivalent of binary bits), providing a mathematical basis for them through the use of ket notation of vector-states. The authors gradually widen in scope to provide explanation of the implications of quantum entanglement and superposition on multiple qubits, as well as providing insight into the problem of measurement. Whilst entangling states into qubits is simple, the paper demonstrates how we can only get one state out of a quantum computer, reducing its potential given the large number of states we can program into it through superposition. The authors give a demonstration of how probabilities are used to skew the likelihood of a useful outcome in our favour. Being a paper targeted at non-physicists, the authors provide concise explanations of the underlying quantum phenomena which are relevant to quantum computers, however these explanations often require the layman reader to assume many facts as a given, giving a shallower depth of understanding on the underlying physics. The paper later provides the reader with insight into how a quantum computer could work, giving a mathematical basis for quantum transformation through linear matrix transformations, which act as quantum logic gates. The notation for these logic gates are also introduced, as well as some example gates like the Cnot gate and the Hadamard transformation. The paper then continues by describing quantum algorithms like Shor's factor algorithm, which hold little relevance to this paper.

Overall, the paper has proven most useful as a source of understanding as to where the massive computational benefits of quantum computing are derived from. Properties of quantum computers like quantum parallelism and the linear scaling of otherwise exponential processor requirement increases due to a superposition of output states are

given with the nuance of their drawbacks, for example, the limitations surrounding measurement of a quantum system. However, the paper only describes how quantum computers work on a theoretical level, as opposed to analysing the technical limitations and viability issues that might occur when constructing one, which reduces the paper's utility. This paper has one major drawback, having been written 23 years ago in September of 2000. Whilst the paper is peer-reviewed, written by reputable authors, and has received a total 178 citations, the study's age is incongruous with the modern quantum computing landscape given what has been a rapidly evolving field over the past 20 years. In spite of this, the fundamental building blocks of how quantum computers operate have changed little, and, when corroborated with excerpts of more recent papers, as a description of how quantum computers work, the paper's information is still largely valid. It is mainly the solutions to the limitations listed in the paper that have evolved, so it follows that descriptions of how quantum computers operate, in terms of the underlying physics, must still be accurate, given the laws of physics have continued to operate as they did 23 years ago. Unfortunately, this does have the implication that the paper's details on how quantum computers are constructed are outdated and thus unusable, though, given the paper seeks only to describe a quantum computer at a theoretical level, rather than an applied, technical one, the more abstract and theoretical facets of the paper do stand the test of time.

2.3 Present landscape of quantum computing

Hassija, V., Chamola, V., Saxena, V., Chanana, V., Parashari, P., Mumtaz, S. and Guizani, M. (2020), Present landscape of quantum computing. IET Quantum Commun., 1: 42–48. <https://doi.org/10.1049/iet-qtc.2020.0027>

This paper begins with a concise overview of how quantum computers operate, how they're mathematically represented and, briefly, how entanglement and superposition make quantum computers so powerful. It then goes on to outline 4 key regions where quantum computers can demonstrate a clear advantage over their classical counterparts: combinatorial problems, linear algebra problems, differential equation problems and factorisation. Combinatorial problems involve optimisation problems, such as finding the shortest distance amongst a given set of points, which the paper claims are not tractable for conventional classical computers via brute force search. The paper states that linear algebraic problems, prominent in emerging fields like machine learning, could be a region where quantum computing excels. The authors place simulation applications under the category of problems involving differential equations, which they claim quantum computers will be suited well to solve. Under factorisation problems, the authors raise concerns that current encryption methods, reliant on the decomposition of large primes, may be vulnerable to quantum computers. This worry is enumerated in the paper's next section, where it also details the various cryptographic methods with which to build quantum-resistant cryptography. However, this paper is most useful as an overview of the quantum

computing hardware landscape. It gives concise overviews of various competing methods with which to build quantum computers, such as superconducting circuits, trapped ions, Nuclear Magnetic Resonance, and topological quantum computing, giving brief descriptions of how they work, as well as which private companies are pursuing them.

This paper comes from a peer reviewed, reputable journal, with 22 citations in 3 years. Written in December of 2020, this paper is beginning to become slightly outdated, given the pace at which the quantum computing field has advanced, however, the information given on applications of quantum computers are still accurate, it is only the details on the paper's contemporary recent advantages that have changed. This paper has proven itself useful as an overview of the quantum computing landscape but falls short of providing adequate explanation and detail on most sections, most notably its incredibly brief overview of how different quantum computers function. The paper is often too concise to be of great value, acting more as a padded list of quantum technologies. This is heightened by the paper's paucity of analysis, for instance claiming that Nuclear Magnetic Resonance quantum computers have fallen out of favour with little information as to why. There are some concerns surrounding the paper's data on publications per year, given the information is sourced from IEEEXplore, which does not contain all quantum related publications, however it could be argued that the overall trend is still representative of the quantum computing field. Despite this, the authors overall have succeeded in providing an overview of quantum computing, the competing technologies, and which private corporations are developing a stake within the field.

2.4 Forthcoming applications of quantum computing: Peeking into the future

Hassija, V., Chamola, V., Goyal, A., Kanhere, S.S. and Guizani, N. (2020), Forthcoming applications of quantum computing: peeking into the future. IET Quantum Commun., 1: 35–41. <https://doi.org/10.1049/iet-qtc.2020.0026>

This paper's stated aim is to summarise the possible future applications of quantum computers, detailing their uses in terms of 4 main categories: Quantum algorithms, Quantum machine learning, quantum simulation, and miscellaneous quantum applications. Beginning with the applications of various quantum algorithms, it details Shor's factor algorithm and its uses for cryptography, as well as Grover's algorithm which can be used to sort unstructured data, which the authors claim is a significant portion of the total data generated. Furthermore, it demonstrates the use of these various algorithms for searching, pattern matching as well as various other useful functions. Most algorithms the authors provide serve as vital parts of a variety of different applications, serving as building blocks for machine learning, optimisation problems, quantum simulation and others. It is these

applications that the paper details next. It provides a useful description of the potential benefits of quantum computing in machine learning, detailing specific algorithms that are used in machine learning like Hidden Markov Models which quantum computers may have an advantage over their classical counterparts. The next section provides the reader with a brief explanation of quantum simulation, demonstrating how quantum computers can, through their inherent quantum nature, simulate nature itself much more efficiently. The last sections detail logistics and financial risk analysis, uses quantum computing to speed up and enhance Monte-Carlo financial simulations to assess Value at Risk in finance, as well as using quantum methods to solve optimisation problems like logistics and supply chains.

This paper is of analytical limited use given its brevity and lack of depth in explanation, serving as a launch pad for future avenues of research. Whilst useful as a summary, it pales in comparison to the utility of other similar papers. However, this paper is peer reviewed from a respectable journal with 11 citations in one year, which suggests it has proven useful as a reference for other papers, further implying at a high degree of accuracy of stated facts. Thus, this paper will predominantly be used to corroborate facts from older sources, as well as those that may have received fewer citations.

2.5 Commercial applications of quantum computing

Bova, F., Goldfarb, A. & Melko, R.G. Commercial applications of quantum computing. *EPJ Quantum Technol.* 8, 2 (2021).

<https://doi.org/10.1140/epjqt/s40507-021-00091-1>

The key thesis of this paper is the overall application of quantum computing on complex combinatorics problems. The paper begins with an introduction of combinatorics problems, briefly explaining how quantum computers may be of help. It outlines 4 key areas where quantum computers may have advantages over their classical counterparts, specifically with the combinatorics problems that relate to each field. The authors begin with a description of quantum cybersecurity, detailing the advances made, and the potential pitfalls if we allow the ability of quantum computers to overtake the difficulty of our foremost encryption algorithms. The paper goes on to inform the reader about the benefits in chemical engineering that quantum computing could bring. The authors describe how, through quantum simulations, computational chemistry can be greatly enhanced, taking the serendipity out of drug and materials discovery.

The paper then continues with analysis of the potential impacts that quantum computation could have on banking and finance, detailing the multitude of combinatorics problems, such as credit scoring that could be greatly enhanced by quantum computers. The paper suggests that, given the potential profitability of developing quantum finance computation, this area may rapidly progress in the current to near term future. To finish, the authors give an

application of quantum computing on the combinatorics problems associated with advanced manufacturing.

Overall, the authors provide the reader with a wealth of analytical information, detailing not only the combinatorial problems within each field, but also the current state of progress in each area, including the companies currently working towards solutions to the given problems. Given this paper was authored in 2021, it is reasonable to assume that this information is not outdated, particularly when corroborated with other sources. However, whilst the authors provide detailed descriptions of each section, they produce little explanation as to why, which hinders any attempt at deeper analysis of this paper. Despite this, being from a peer reviewed journal, with 50 citations in under 3 years, the facts given in the paper can be taken as true with a high degree of confidence.

3 The building blocks of quantum computers

In order to fully analyse and evaluate the merits and viability of quantum computing, it is crucial to first understand where the quantum benefits arise from, and where exactly the challenges of constructing a functional, noise-less quantum computer lie.

3.1 The quantum computational process

The vast majority of viable quantum computers operate with the three-step process illustrated in Figure 1.

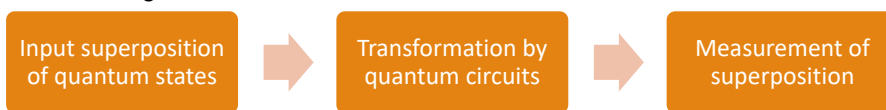


Figure 1 – the three key stages of quantum computation.

These will be discussed in order of increasing abstraction, from the superposition of qubit states – involving qubits and their unique properties, through measurement of superposition and its many crucial implications, to quantum circuits - which serve to transform and manipulate qubits.

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3.2 Qubits and superposition

Whilst quantum computing's classical counterpart utilises binary "bits" to perform operations, each either in an ON or OFF state, quantum computers operate on quantum bits, known as qubits (Gil', et al., 2023). There lie two key differences between qubits and classical bits - the ability of qubits to exist in states of superposition and the ability for multiple qubits to become entangled together (Hassija, et al., 2020).

Superposition permits qubits to exist in multiple states simultaneously, thus whilst a classical bit can be represented by a binary number, 0 for OFF, and 1 for ON, qubits must be represented by a unit vector in a region of possible quantum states. The state of a qubit is therefore defined as the weighted superposition of two states 0 and 1, $a|0\rangle + b|1\rangle$, where a and b are complex numbers such that $|a|^2 + |b|^2 = 1$ and the states $|0\rangle$ and $|1\rangle$ are orthogonal unit basis vectors. (Rieffel & Polak, 2000) This means that, whilst classical bits can exist only in 2 states, qubits can exist in an infinite number of states.

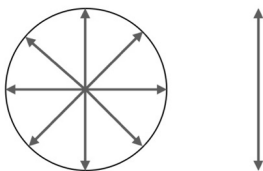


Figure 2 - the region of possible values a qubit could exist in, and the region of possible values a binary bit could exist in

3.3 Measurement

Whilst a qubit in superposition, with, for example, a spin state $|\psi\rangle = a|\uparrow\rangle + b|\rightarrow\rangle$ (where spin is taken as the computational state of the qubit) can exist in an infinite number of possible states, when measured, the qubit's superposition collapses into one of two states. This means that the qubit's spin will, when measured, be projected onto either $|\uparrow\rangle$ or $|\rightarrow\rangle$ basis vector, so the probability of the electron collapsing into each state is given by the magnitude of each weighting squared ($|a|^2 = P(|\uparrow\rangle)$ and $|b|^2 = P(|\rightarrow\rangle)$). Crucially, after measurement, it is impossible to determine what the initial state of $|\psi\rangle$ was, which significantly limits the benefits quantum computers may have over their classical counterparts – this will be discussed in greater detail under the section *The utility of measurement*. (Rieffel & Polak, 2000)

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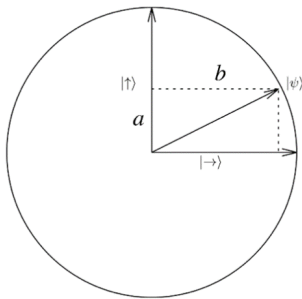


Figure 3 - depicts the projection of some state $|\psi\rangle$ onto the basis vectors (Rieffel & Polak, 2000)

Note that these basis vectors are not intrinsic to the electron, but rather defined by the measurement device.

3.4 Multiple qubits and entanglement

Measurement of a system of two qubits can be treated as measuring a system with 4 orthonormal basis vectors ($|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$), for each qubit contributes its own set of basis vectors. (Rieffel & Polak, 2000)

Quantum entanglement is a quantum phenomenon whereby two different quantum particles in superposition are linked such that the measurement of one can determine the measurement outcome of the other. For example, the state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) + 0|01\rangle + 0|10\rangle$ is entangled as, if the first qubit is measured as $|0\rangle$ the second qubit can only be measured as $|0\rangle$ (as the only possible basis state beginning with $|0\rangle$, $|00\rangle$, also ends with $|0\rangle$). However, for example, the state $\frac{1}{\sqrt{2}}(|10\rangle + |11\rangle)$ is not entangled, for, if the first qubit is measured as $|1\rangle$ the second qubit could collapse into either $|0\rangle$ or $|1\rangle$. This has the crucial implication that there are states that a quantum computer can be set in, that cannot be described in terms of individual qubits. (Chamola, et al., 2020)

3.5 Quantum Scaling

With quantum computing's classical counterparts, in order to describe the state of a system of 3 bits, 6 numbers are required. For example, to describe the state 101, a set of place value positions, and state values for each bit are required. i.e.

1. Position 1:
2. Value 0
3. Position 2:
4. Value 1
5. Position 3:
6. Value 0

To generalise, for n bits, there are $2n$ possible states that can be described. (Rieffel & Polak, 2000)

However, due to the fact that some quantum states cannot be described in terms of individual qubits, the state of a quantum system can be described only in terms of its possible output values – the orthonormal basis vectors. This means that, for 3 qubits a quantum computer can encode 8 pieces of information: $a|000\rangle + b|001\rangle + c|010\rangle + d|011\rangle + e|100\rangle + f|101\rangle + g|110\rangle + h|111\rangle$. This ultimately means that, for a system with n qubits, a quantum computer can describe 2^n states. Thus, whilst the computing power of classical computers scale linearly, quantum computers scale exponentially. (Chamola, et al., 2020)

3.6 Quantum parallelism

The ability of a system of multiple qubits to exist in a large number of states simultaneously through superposition has powerful implications. Applying a function to the superposition of states will transform all the superposed states simultaneously at no extra computing cost. This property of quantum computers is known as *quantum parallelism*, and, when exploited fully, serves as a key aspect of any theorised quantum supremacy. (Rieffel & Polak, 2000)

However, there exists a major caveat, though quantum computers can simulate multiple states simultaneously, they can output only the result of one of those states. This is derived from the property of measurement that renders it physically impossible to recover the initial superposed states post-collapse, resulting in only the one collapsed output. (Gil', et al., 2023) Furthermore, this output is picked probabilistically (based off the magnitude of the projection of each state onto the basis vectors squared – for example, $|a|^2$), so often it is not certain that the desired result will be outputted. Consequentially vast amounts of research and development have been invested in developing a quantum computer that reliably outputs the useful information. (Rieffel & Polak, 2000)

3.7 Quantum Logic Gates and Circuits

Whilst classical computers use logic gates to perform operations on bits, quantum computers use quantum state transformations, often referred to as quantum gates.

3.7.1 Basic quantum gates

The transformations of qubits, represented as vectors, are performed through the use of unitary transformations. These transformations may be described by use of a matrix.

$$\begin{aligned}
 I: & \begin{array}{l} |0\rangle \rightarrow |0\rangle \\ |1\rangle \rightarrow |1\rangle \end{array} & \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\
 X: & \begin{array}{l} |0\rangle \rightarrow |1\rangle \\ |1\rangle \rightarrow |0\rangle \end{array} & \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\
 Y: & \begin{array}{l} |0\rangle \rightarrow -|1\rangle \\ |1\rangle \rightarrow |0\rangle \end{array} & \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \\
 Z: & \begin{array}{l} |0\rangle \rightarrow |0\rangle \\ |1\rangle \rightarrow -|1\rangle \end{array} & \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}
 \end{aligned}$$

The given names are a convention, I refers to the identity matrix, X refers to negation, Z refers to phase shift and $Y = XZ$. Collectively, the XYZ gates are referred to as the Pauli gates and represent rotations around each axis by π radians. (Rieffel & Polak, 2000)

Diagrammatically, simple quantum transformations are represented as follows:

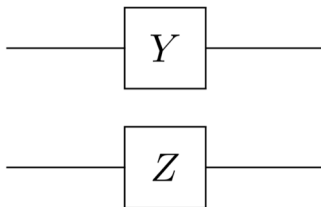


Figure 4 - Graphical representation of Y and Z quantum logic gates. (Rieffel & Polak, 2000)

3.7.2 The C_{not} Gate

Quantum computers with multiple qubits can take advantage comparison gates, such as the controlled not, or C_{not} gate, which inverts the value of the second qubit if the first qubit is measured as 1. (Rieffel & Polak, 2000)

$$C_{not}: \begin{array}{l} |00\rangle \rightarrow |00\rangle \\ |01\rangle \rightarrow |01\rangle \\ |10\rangle \rightarrow |11\rangle \\ |11\rangle \rightarrow |10\rangle \end{array} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Graphically this may be represented as follows, where a circle represents the control qubit, and a cross represents negation conditional upon the control:

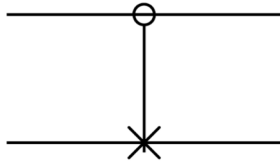


Figure 5 - Graphical representation of a Controlled Not gate (Rieffel & Polak, 2000).

3.7.3 The Hadamard Gate.

The Hadamard gate generates superposition states, mapping the basis states into an equal superposition.

$$H: \begin{aligned} |0\rangle &\rightarrow \frac{|0\rangle + |1\rangle}{\sqrt{2}} \\ |1\rangle &\rightarrow \frac{|0\rangle - |1\rangle}{\sqrt{2}} \end{aligned} \quad \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

This gate is of great importance to quantum computing, for the superpositions it generates underly the uses and benefits that quantum computers present. (Rieffel & Polak, 2000)

3.7.4 The Toffoli gate

An extension of this is the Toffoli gate, which requires 3 qubits, negating the third qubit if both the first and second qubits are 1. (Rieffel & Polak, 2000)

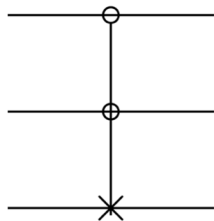


Figure 6 - a Toffoli gate (Rieffel & Polak, 2000).

Furthermore, it can be shown that the Toffoli gate can be decomposed as a circuit of multiple smaller gates.

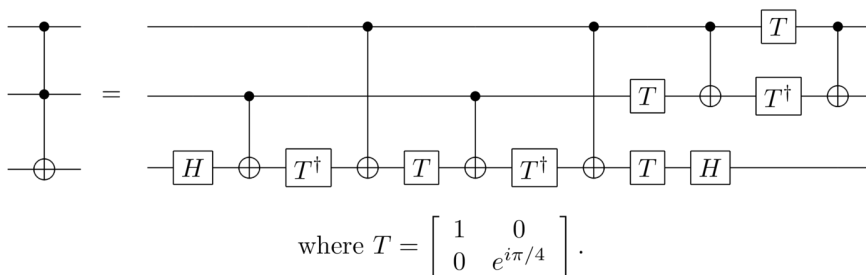


Figure 7 - Decomposition of a Toffoli gate (Wikimedia, 2019).

3.8 Quantum Circuits

A sequence of quantum transformations, known as a quantum circuit, act in similar ways to their classical counterparts, for example, a one-bit adder with two data qubits $|a\rangle$ and $|b\rangle$ and 2 qubit output $s = s_0 + s_1$ (representing the sum of $|x\rangle$ and $|y\rangle$), can be denoted as a *Toffoli* gate followed by a C_{not} gate.

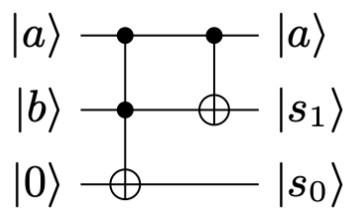


Figure 8 - graphical representation of a quantum one bit adder (Rieffel & Polak, 2000)

With quantum circuits, it is important to note that each line, known as *quantum wire*, refers to a specific qubit, thus each operation on a line operates on the qubit it represents. (Rieffel & Polak, 2000)

3.8.1 Application of classical algorithms on quantum circuitry.

Quantum logic gates, by virtue of their underlying unitary transformations, are always reversible, unlike many key classical gates (Rieffel & Polak, 2000). However, quantum computers can still perform any classical computational algorithm despite having access only to reversible gates. For example, the Toffoli gate can perform all irreversible Boolean operators.

Crucially, though quantum computers can perform any classical algorithm, they are unable to do so any more efficiently or reliably than classical computers can (Chamola, et al., 2020). Thus, achieving a quantum advantage is not simply a case of applying pre-existing classical algorithms to a faster computer, but rather a paradigm shift in the computation itself.

3.9 Summary

In order to achieve quantum supremacy, any quantum algorithm must rely upon the quantum phenomena, such as superposition and entanglement, that underpin how quantum computers function, including taking full advantage of the emergent properties such as quantum parallelism and exponential scaling that set quantum computers apart from their classical alternatives. As a result of this, it is the author's belief that quantum computers are unlikely to replace classical computers, but rather carve their own niche in regions that classical computers cannot currently venture.

4 Challenges of constructing quantum computers

There are a multitude of barriers inhibiting the development of quantum computers which must be overcome before quantum computers can viably hope to challenge classical computation.

4.1 Decoherence

The most prominent of issues plaguing quantum computation is the problem of decoherence, with quantum computers susceptible to decoherence generally described as “noisy” (Preskill, 2018). States of superposition are highly fragile and are susceptible to losing their coherent properties when interacting with the environment (Preskill, 2018). This results in the premature collapse of any superposition and leakage of information to the environment, which physically cannot ever be recovered, limiting the time that a quantum computer can functionally perform operations on qubits (Chamola, et al., 2020). Furthermore, decoherence results in a reduction in qubit scaling efficiency, for the necessary pre-emption of decoherence and resulting assumption that not all qubits will remain functional produces quantum computers designed with a greater number of qubits than necessarily required. It is not currently the case that the main difficulty in constructing quantum computers lies in increasing the qubit count, but rather in increasing the coherent qubit count (Gil, et al., 2023). Therefore, in order to combat this issue, quantum computers must be kept as near to absolute zero as possible. At 0.015 Kelvin, a quantum computer is the coldest known place in the universe, thus simply maintaining such temperatures requires vast level of complexity and energy.

4.2 Error correction

Efforts in combating system decoherence have largely relied not upon hardware improvements but rather efficient error correction protocols and other similar mitigation techniques (Gil, et al., 2023). Unlike classical computers, which have discrete errors, the variable probabilistic nature of quantum computing result in continuous errors, which involve both amplitude and phase (Gil, et al., 2023). Furthermore, the inability to copy quantum states, as result of the no-cloning principle, and the property of measurement that collapses quantum states, destroying superposed information in the process, means that highly developed and efficient classical error correction algorithms cannot be applied to their quantum counterparts - novel methods of correction must be sourced.

4.3 Utility of measurement

Once the quantum state of a system has been measured, it is impossible to determine the initial superposed states. Furthermore, due to the no cloning principle, a qubit's state cannot be copied and measured multiple times to unveil more information (Rieffel & Polak, 2000). This greatly reduces the theoretical advantage that quantum computers may have, greatly

restricting the power of quantum parallelism. Thus, in order to ensure measurements are useful, the transformations of qubits through logic gates and circuitry must influence the probabilities in such a way that the desired solutions are outputted, which further limits the range and increases the complexity of quantum algorithms.

5 Overview of competing methods of quantum computer construction

There are many different methods by which quantum computers may be constructed, each with different advantages, and in differing stages of development. The main differentiating factor between quantum computers is their choice of qubit.

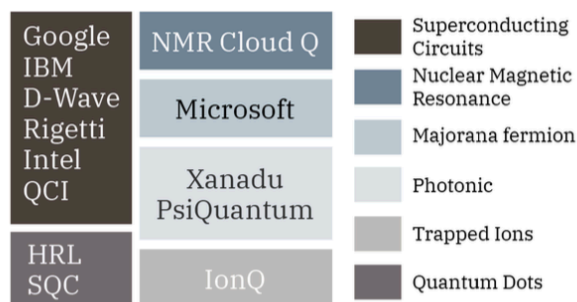


Figure 9 - summary of leading quantum computer companies categorised by qubit choice. (Chamola, et al., 2020)

Photonic qubits have the advantage of remaining coherent for relatively long periods of time, with coherence times of $\sim 150\mu s$ (Gil', et al., 2023). Furthermore, photons have the ability to operate at room temperature, which significantly reduces energy consumption. However, constructing circuits for photon-based qubits presents greater challenges than other qubit choices (Gil', et al., 2023). Silicon qubits have great theoretical stability, with coherent times ranging from $\sim 1 - 10s$, but require stronger error correction schemes due to a significantly worse gate fidelity (Chamola, et al., 2020).

Topological quantum computing is a method which aims to drastically reduce errors by encoding information in an object's topology, properties preserved under continuous deformations like twisting. (Chamola, et al., 2020) Since topological quantum computers are interested only in topology, they are much less error prone, with an expected gate fidelity of $\sim 99.9999\%$, and as such show great promise. However, this technology is very much in its infancy, despite being pursued by Microsoft (Chamola, et al., 2020).

Trapped ion qubits are charged ions trapped in an electromagnetic field, where the qubit's computational state is given by its spin energy level (Gil', et al., 2023). Trapped ions have the

long coherence times of ~ 50 s as well as a high gate fidelity of $\sim 99\%$, however, trapped ions lack scalability (Gil', et al., 2023). Quantum computers constructed from trapped ions are notably good at performing high level simulations, such as cosmology and high energy physics (Crane, et al., 2017). Quantum computers based off trapped ions are currently being built by a start-up known as IonQ.

Using superconducting qubits is the current leading method of quantum computing, being actively pursued by large corporations like Google, IBM, and intel, as well as prominent start-ups including D-Wave and Rigetti (Chamola, et al., 2020). Superconductors have no resistance, which theoretically means qubits should remain error free. However, they consequentially require extremely low temperatures below 1 Kelvin, (Gil', et al., 2023) temperatures which further requires large amounts of energy to sustain. Superconducting qubits are amongst the most scalable and are currently the most advanced method of constructing qubits, and thus, in the author's view, are the most viable methods by which a high-impacting quantum computer could be constructed, if not simply due to the large research investment put towards its development.

6 Applications of quantum computers

The impact of a tool can only be measured in terms of its applications. In this sense, quantum computers have the potential to become amongst the most powerful computational tools ever developed. However, it is not simply just the direct impacts of quantum computers that give them their high potential, but rather the quantum computing enabled technologies that the advent of quantum computation may see developed. Just as the inventors of the classical computer may not have possibly conceived the invention of the internet, and the subsequent online revolution, there are doubtless revolutionary quantum-enabled technologies that remain unconceived. Thus, there are 3 main classifications of quantum computational applications to discuss:

Type 1:	• Direct benefits of quantum computation
Type 2:	• Quantum enabled <i>classical</i> technologies
Type 3:	• Quantum enabled <i>novel</i> technologies

6.1 Type 1 applications: Direct benefits of quantum computation

Type 1 applications are applications resulting directly from the superior ability of quantum computers to perform certain tasks. These stem from the two main quantum computational

benefits: quantum scaling and quantum parallelism. Quantum computers are, at a fundamental level, inherently more capable than their classical counterparts at three main types of tasks (Bova, et al., 2021):

- 1. Combinatorics problems
- 2. Searching and sorting algorithms
- 3. Exact models

6.1.1 Combinatorics problems

The aim of combinatorics problems is to figure out the number of possible combinations, or the best possible combinations in some metric of a set of options (Chamola, et al., 2020). An example of this would be the travelling salesman problem, where a salesman needs to visit n number of cities in as short a distance as possible, in one loop (Karthik Srinivasan, et al., 2018). Though initially trivial for a classical computer, the complexity of such problems grows exponentially as n increases (Karthik Srinivasan, et al., 2018). Therefore, for large problems analogous to the traveling salesman, room sized classical supercomputers are required (Bova, et al., 2021). Quantum computers however are capable of processing large volumes of data at the same time through quantum parallelism, permitting them to solve large and highly complex problems with the collapse of a superposition.

6.1.2 Searching and sorting algorithms

The ability to rapidly search and sort large volumes of data is fundamental to any computational algorithm, from machine learning to international supply chains (Gil', et al., 2023). Through Grover’s algorithm, quantum computers are significantly faster than their classical counterparts at these tasks, with an improvement by a quadratic factor over classical computing (Figure 10) (Hassija, et al., 2020).

Application	Classical complexity	Quantum complexity
finding minimum of an unordered list	$O(N)$	$O(\sqrt{N})$
determine graph connectivity	$O(N^2)$	$O(N^{3/2})$
pattern matching	$O(n + m)$	$O(\sqrt{n} + \sqrt{m})$

Figure 10 - Comparison of speed-ups achieved due to Grover's algorithm. (Hassija, et al., 2020)

As searching and sorting are fundamental to most computational algorithms, any quantum translation of current classical algorithms that utilises Grover’s algorithm (or some variant of it) is by design likely to supersede the original (Hassija, et al., 2020). These vast benefits stem from quantum parallelism, permitting quantum computers to process data in parallel, in addition to quantum scaling, which exponentially increases the power of quantum parallelism by permitting more data to be processed simultaneously.

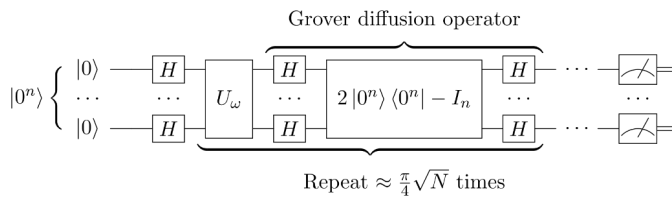


Figure 11 - Quantum circuit of Grover's algorithm. (Fawly, 2021)

6.1.3 Exact models

The original conception of quantum computation by Richard Feynman in 1981 was as a means to simulate nature exactly (Feynman, 1986). Whilst classical computers must necessarily reduce any complex simulation down to discrete binary bits, which result in approximations and the omission of fine details, leading to the propagation of errors, quantum computers, by virtue of their inherent continuous quantum nature, are theoretically able to model nature exactly (Hassija, et al., 2020). This has the potential to result in many major advancements in a multitude of fields, and also is the root source of many *Type-2* applications' benefits. In order to generate exact models capable of simulating quantum phenomena beyond the reach of classical computers, small scale systems ranging from only (50-100) coherent qubits are required, known as quantum simulators (Gil', et al., 2023). The author views these simulators as a highly viable stepping stone for the development of quantum computers, allowing quantum computation to have a large impact at a relatively early stage of their development.

6.2 Type 2 applications: Quantum enabled classical technologies.

Quantum enabled classical technologies are algorithms and process designed around classical computers that have been adapted for and enhanced by quantum computers. These applications utilise quantum computers in (Karthik Srinivasan, et al., 2018) conjunction to, or as a replacement for classical computers.

6.2.1 Quantum biochemical engineering: High Throughput Screening

Currently, drug development is a long, drawn-out process of trial and error, requiring billions of dollars in expenditure, and often taking over a decade for a single drug (Cao, et al., 2018). As such, un-subsidised drug costs are high, and many potentially curable diseases deprioritised in favour of curing higher reward diseases. Thus, if quantum computers can disrupt this industry and improve both drug efficacy and streamline the research process, there is a potential to save a significant number of lives. However, in order to analyse any

potential benefit quantum computers may provide to the pharmaceutical industry, it is important to first understand where the time and costs are most heavily invested. (Cao, et al., 2018)

The current most common method of drug discovery is high-throughput screening, where a large molecular library is screened through *in vitro* experiments. These typically follow a 5-stage process (Cao, et al., 2018):

1. Generate library of potential drug candidates.
2. Screen based on medicinal activity to identify chemicals known as hits that exhibit desired medicinal activity against a disease.
3. Screen further based on factors like efficacy and potency of hits.
4. Optimise into candidates called lead compounds.
5. Proceed through stages of drug development and clinical trials till one lead becomes a viable commercial product.

Of this process, a significant portion of time and money is invested in manufacturing and identifying hits, which requires the synthesis of a large number of compounds (Cao, et al., 2018). Whilst classical computers, introduced in the 1970s, have increased hit rates and lead quality, they can only accurately simulate molecules of low molecular weight (less than 900 Daltons), resulting in 78% of the current drugs market being of low molecular weight (Cao, et al., 2018).

By simulating the ability of a potential drug to bind to a biological target, computers can streamline the discovery process, however, quantum computers, through quantum parallelism and exact models, are able to more accurately and rapidly simulate the complex chemistry required to screen possible hits (Cao, et al., 2018).

6.2.2 Numerical weather prediction

Weather simulations models are highly complex, with the chaotic nature of weather patterns being a significant challenge to classical computers (Gil', et al., 2023). Problems such as the most likely path of a tropical cyclone require vast, power hungry, classical supercomputers to find near-accurate approximate solutions (Gil', et al., 2023). As a result, current weather prediction models, though reasonably accurate, are limited by the inherent discrete nature of classical bits, for any set of continuous data must be reduced to a binary ones and zeros (Gil', et al., 2023). Furthermore, the computational capacity of classical bits is far too minimal to produce sufficiently accurate predictions into the future. However, quantum computers do not face similar limitations, for qubits are inherently continuous, which allows quantum computers to take less approximations than classical computers, and the quantum scaling of quantum computers, combined with the power of quantum parallelism allows quantum computers to perform more powerful simulations. Though these improvements are rooted in the ability of quantum computers to perform exact models, it

should be noted that quantum computers will not be able to simulate weather exactly, rather they shall provide dramatic gains in reliability and prediction length (Gil', et al., 2023).

The possible impacts of improved numerical weather predictions are vast, for with improved weather prediction capabilities comes the ability to evacuate and warn people in the path of storms. Furthermore, improvements to weather models can help simulate flood surges, which allows governments to better prepare in advance with reinforced embankments and higher densities of cyclone or tornado shelters.

Factors	TC 1970	TC 1985	TC 1991	TC 1998	TCs 2007– 2009	TCs 2013– 2016	TCs 2017– 2020
1. Lack of effective warning system by the GoB	338 ^a	319	199	88	34	13	–
2. Lack of embankment or not repairing embankment	337	246	249	124	59	24	15
3. No or limited cyclone shelters	336	279	215	126	78	28	14
4. The residents did not believe in warning	324	305	255	56	34	16	14
5. Limited volunteers, microphone and loudspeaker to disseminate warning during extreme weather	322	256	245	112	44	23	12
6. Kutcha houses including mud-made houses and their collapse caused people's deaths	327	262	217	131	62	35	31
7. Kutcha road networks	332	272	215	124	55	17	15
8. Limited availability of devices (radio, TV, mobile) to receive warning in the past	316	262	188	57	35	8	–
9. Limited strong building or other type of infrastructure nearby	307	267	265	121	22	9	3

Figure 12 - Factors contributing to deaths in tropical cyclones within Bangladesh (GoB = government of Bangladesh). (Alam, 2023)

As shown in figure 12, factors relating to unpreparedness, such as the lack of evacuations, embankment construction and weak houses have resulted in a large loss of life (Alam, 2023). Though Bangladesh has seen great success in decreasing the fatality rates tropical cyclones inflict, the onset of climate change, and the resulting unpredictability and increased power of cyclones, will likely hinder efforts to reduce these losses to 0 (Alam, 2023). Furthermore, Bangladesh's drop in fatality rate has been largely the result of a reactive approach, re-building stronger in hard hit areas and experimenting with different mitigation techniques to minimises deaths. (Alam, 2023) However, with quantum computational simulations, a pro-active approach may be employed, studying where cyclones are likely to hit and instigating preventative measures before any disaster.

6.2.3 Quantum assisted Artificial Intelligence.

The ability of quantum computers to handle large datasets efficiently has great potential in the field of Artificial Intelligence (AI). In order to train an advanced AI, the processing of large

datasets are required. This requires time and electrical energy, which in of itself is costly (Rawat, et al., 2022). Furthermore, the great amounts of energy required to power the GPUs that AI's train on is immense, with ChatGPT-3 estimated to have contributed over 500 metric tons of CO_2 , equivalent to 600 New York to London flights (Heikkilä, 2022). This large cost is due to the relative inefficiency of classical computers at computing large volumes of data. Quantum computers, however, through quantum parallelism enhanced by quantum scaling, can easily handle data sets much more efficiently than classical computers. Furthermore, the training itself would be greatly enhanced by the addition of quantum searching and sorting, allowing larger datasets and unsorted datasets to be processed effectively. In addition, quantum computers are well optimised to deal with high-dimensional datasets, for entangled qubits themselves operate in a manner analogous to higher dimensional vector spaces (Ying, 2010). Performance of classical computers suffers when applied to these datasets, providing quantum computers a clear advantage (Rawat, et al., 2022). Though quantum computers in the order of 1000 coherent qubits are required for any substantial benefit over current classical supercomputers, progress has been made towards merging AI with quantum computation. For example, a state of the art "meaning aware" natural language algorithm, one that understand both the whole sentence holistically, as well as the meaning of each individual word, has been successfully implemented on a quantum computer (Rawat, et al., 2022). In addition, TensorFlow, a popular machine learning library developed by Google, has implemented quantum computational functions, both improving accessibility towards development of quantum AI algorithms through its greater abstraction, and increasing the platform and perception of quantum computers amongst the computer science community as an emerging technology no longer in its infancy (Google, 2020). This supports the belief that quantum computers will have a viable impact upon AI in the future.

6.2.4 Quantum Banking and Finance

Combinatorics problems are common withing banking and finance, ranging from arbitrage to credit scoring and risk assessment (Bova, et al., 2021). If these problems grow too complex, which occurs often due to the exponential scaling of combinatorics problems, classical systems have to resort to estimation and Monte Carlo simulations in order to generate near accurate solutions (Bova, et al., 2021). However, quantum computers, by virtue of parallelism and exponential scaling would be able to process much larger problems than our current classical computers can, and at much lower energy cost. However, a McKinsey report has stated that many banks have chosen to reduce the use of classical Monte Carlo simulations for value-at-risk calculations, which stems from the unreliability of such simulations when performed quickly (EGGER, et al., 2020). This is because running Monte Carlo simulations on classical computers is incredibly time consuming. However, this is not the case with quantum computers, which are able to solve Monte Carlo system much faster (EGGER, et al., 2020).

6.3 Type 3 applications: Quantum enabled novel technologies.

Quantum enabled novel technologies are applications of quantum computers that have no classical equivalent. Designed for, and not possible without, quantum computers, they represent a paradigm shift in approach to computing, being designed from the ground up to take advantage and utilise the quantum nature of quantum computers in a way not possible to achieve by classical computers. These are the hardest to predict given the relatively early stage of quantum computer development. However, the applications within this category are those that the authors views as possessing the greatest potential to be of high impact.

6.3.1 Quantum internet

A quantum internet utilises quantum phenomena to send information, stored in qubits, over long distances with absolute security. The security of a quantum internet relies upon a technique known as Quantum Key Distribution. If one party, sends an entangled qubit, retaining the other of the pair, to another party, who, upon receiving the qubit, measures it, the measurements of both parties can be freely compared over a public channel without any threat to the encoded data, as they do not individually reveal the original message (Manojlovic, 2017). This also means, if they do not match, that the entanglement has collapsed, and someone has intruded. But as the individual decohered qubits are of no help, the interceptor cannot decipher any viable message (Ananthaswamy, 2019).

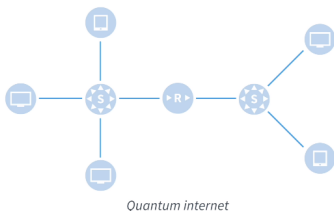


Figure 13 - Diagram of a quantum internet. (QuTech, n.d.)

The network is comprised of quantum computer end-nodes. The switch directs single qubits along the system which get sent to a quantum repeater, which act as a range extender for the entangled qubit, as sending qubits over long distances will increase the chance of losing information to decoherence (Wehner, 2018). A benefit of the quantum internet is that wires used do not have to be re-laid, for entangled photons, used as qubits, can be sent down existing optical fibres with little loss of reliability. This allows an easier transition to a quantum internet, using the existing infrastructure.

However, quantum error correction schemes capable of protecting qubits would require a larger number of qubits, and so, a larger quantum computer capable of coding and decoding the information. Redundancy, contrary to classical bits, cannot be built into a quantum

system due to the no cloning theorem. There are, however, some codes that can protect against redundant qubits, for example Shor code. These work by distributing information across multiple qubits so operation errors and loss errors can be corrected (QuTech). Quantum internets have yet to leave the laboratory but scaled tests have occurred. There have also been successful networks in Geneva (CERN to Geneva university), a setup in Tokyo in partnership with the EU, which, for the first time, implemented a higher-data rate encryption capable of supporting secure voice and video conferencing.

6.3.2 Quantum biochemical engineering: Computer Aided Drug Design

Quantum computers open up exciting *Type 3* possibilities within the field of biochemical engineering that remain beyond the reach of classical computation. Whilst high throughput screening is dependent upon (computer aided) luck to find molecules that exhibit certain desired traits, computer aided drug design instead seeks to do the reverse (Cao, et al., 2018). From the desired therapeutic effects, computer aided drug design aims to engineer the perfect chemical structure that exhibits such characteristics. If made reliable and commercialised, computer aided drug design has the potential to completely revolutionise drug development, taking the serendipity out of a decades-long laborious and expensive process (Cao, et al., 2018). Though there have been basic trials to perform computer aided drug design on classical computation, the introduction of quantum computers, with their ability for exact simulations and greater processing power and speed, would result in a complete paradigm shift in drug manufacturing, allowing more drugs to be created targeting a wider variety of diseases, from common high-impact targets to the more obscure, at a faster rate for a cheaper price (Cao, et al., 2018).

7 Growth of quantum computing

Quantum computing is a rapidly developing field, given both the potential high impact of a successful quantum computer, and the lucrative benefits that entice the private sector towards its development. Major multinational corporations such as IBM, Google, Microsoft, and Alibaba are actively developing quantum computers, as well as multiple dedicated quantum enterprises such as D-wave and Rigetti. Investment in this area is high, with many governments providing funding, such as the Australian centre for computation and communication technology, the European Union's quantum technologies flagship, or the Chinese quantum national laboratory for quantum information science.

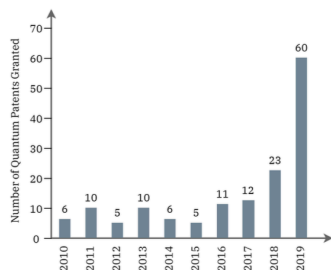


Figure 15 - number of patents granted in the years 2010 – 2019. (Chamola, et al., 2020)

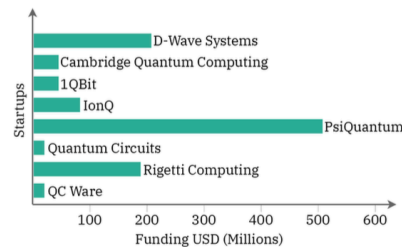


Figure 14 - Funding awarded to quantum computing start-ups. (Chamola, et al., 2020)

As demonstrated in Figures 14 and 15, quantum computing is a healthy and active field, with the quantum computing market expected to reach \$8 Billion by 2027 (Chamola, et al., 2020). For this reason, the author believes that rapid and sustained growth can be expected of the quantum computing field within the next decade.

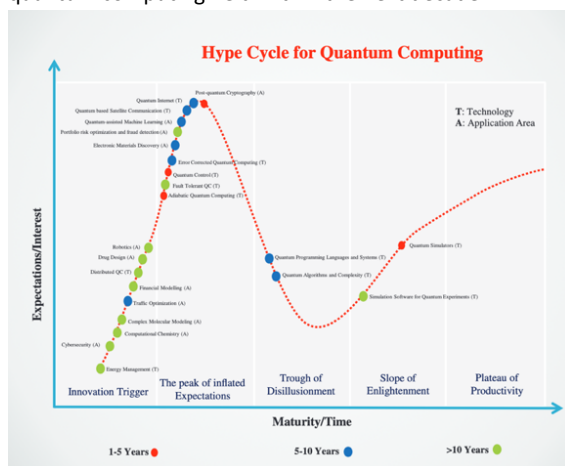


Figure 16 - Hype cycle diagram for quantum computing. (Gil', et al., 2023)

Furthermore, as shown in Figure 16, a large proportion of quantum technologies are on the “crest of interest” within quantum computation’s projected hype cycle, which further supports the view that the next decade may see a boom in investment and growth. Though the trough of disillusionment looms ahead, it is the author’s belief that the potential benefits for quantum computation will keep expectations and interest sufficiently high.

8 Conclusion

This paper concludes that quantum computing is a rapidly developing field, one in which sustained growth is near guaranteed. Quantum computers have incredible potential, with powerful new computational methods like quantum parallelism, resulting from entanglement and the collapse of superpositions. Furthermore, quantum scaling allows quantum computers to scale exponentially in computational ability with a linear increase in qubits, resulting from the “extra” entangled states that classical computers, by means of their discrete bits, cannot use. When combined into larger algorithms, such as Grover’s search algorithm, these benefits permit any algorithm designed for quantum computers to outperform any equivalent classical computer.

However, it is the practical impact of the upcoming introduction of quantum computation within the next decade that shall determine its status as a transformative technology. The author views Type 1 applications as highly feasible, but of limited actual impact, thus they serve instead as stepping stones towards the more lucrative Type 2 applications. Quantum biochemical engineering is a region of great interest due to its possibly huge implications for the drugs industry, and, by extension, humanity as a whole. Development of quantum algorithms aimed at drug development has the potential to save countless lives and rid many more of disease. Numerical weather prediction stands to improve the ability for nations to plan and react to major environmental hazards, which itself could save lives if available at a low cost to developing nations. Quantum computing also has the potential to revolutionise how banking deals with complex combinatorics problems, increasing profit margins. Type 3 applications have the potential to completely transform not just technology, but society itself, with a quantum internet allowing rapid, fully secure communication, and quantum aided drug design representing a paradigm shift in the drugs development process. However, it is likely that the most potent of these applications have yet to be invented.

Though theoretically quantum computers hold great potential, the limitations, and challenges of viably constructing a powerful quantum computer are vast. Decoherence will likely remain a significant issue, with few hardware advancements having been made towards its physical reduction in the *past* decade or intensive research. Software-based error correction is already currently at its limits and will be insufficient in tackling the issue without the development of better hardware methods. Furthermore, the size and energy required to sustain the ultra-low temperatures required for coherence are likely to pose an issue, partially negating the possible benefit of quantum computers for greater profitability or diminished environmental impact with applications such as artificial intelligence and finance. Whilst methods such as topological quantum computing aim to eliminate decoherence, the author does not believe that the technology is advanced enough to have any major impact within the next decade.

Overall, it is the viability of Type 2 applications that the author sees as the deciding factor in the magnitude of quantum computing's impact, given the limited impact of Type 1 applications and the technological limits that will likely inhibit development of any major Type 3 application. Of the major Type 2 applications, the author views those that utilise exact models, such as numerical weather prediction and scientific quantum simulations, as the most viable. This is due to the smaller number of coherent qubits required, the huge limitations on classical computers with simulation tasks that quantum computers do not face, and the likely large volumes of funding and growth resulting from the relatively high impact of such simulations. The smaller qubit requirement reduces the impact of decoherence, and thus the resulting running costs. This opens the possibility that quantum computers may be commercially produced and sold, given the relative stability of such systems and the potential they hold. Whilst current development on quantum computers is focused on building the systems themselves, once quantum computers can be shown as viable products and not just the pursuit of academic research, the perception of quantum computers as a technology in its infancy will be discredited, opening the door to wider recognition and the subsequent development of quantum computing *enabled* applications. Beyond the next decade as quantum computational technology matures, quantum computing will increasingly enter the mainstream domain, as more Type 3 quantum reliant technologies are developed, however whilst research in quantum computing construction is likely to dominate the next decade, development of fully realised applications are unlikely.

The benefits of quantum computers may not be dramatic nor noticeable to any consumer, rather operating behind the scenes, inhabiting in niches where classical computers struggle. However, their impacts will be large, resulting in a multitude of improvements, such as increased scientific knowledge through simulations, lives saved through improved drug treatment techniques, more accurate weather simulations and improved climate analysis. It is for these reasons that the author finds quantum computation as a likely transformative technology within the next decade.

9 Evaluation

The author believes that this dissertation has succeeded in fulfilling its aims of providing a holistic yet through examination on the impacts of quantum computation. However it should be noted that not all impacts and potential technologies could be included within the scope of such paper, and so only applications deemed both viable and impactful enough have been covered in detail. Furthermore, deeper coverage of the different methods of quantum computer construction have been omitted, as they hold little relevance to either understanding the fundamental principles, to which they all adhere, or the resultant benefits, which they all possess. Due to open access tools such as *Google Scholar*, the author has been able to analyse and discuss a wide variety of academic sources and is thus confident that the information reported is representative of the current cutting research edge. If the reader were to continue researching down this region, the author would recommend investigating post quantum cryptography (PQC), which, though beyond scope for this paper, has the potential, if the world remains unprepared, to trigger serious negative implications to online security.

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Project Proposal form

Learner Name	<u>Daniel Fussey</u>	Learner number	<u>1593</u>
Centre Name	<u>Charters School</u>	Centre Number	<u>51529</u>
Teacher Assessor	<u></u>	Date	<u>29/09/2023</u>
Unit	<u>P301</u>		
Proposed project title	<u>To what extent will quantum computers have an impact on computing in the next 10 years</u>		

Section One: Title, objective, responsibilities

Title or working title of project (in the form of a question, commission or design brief)

"To what extent will quantum computers have an impact on computing in the next 10 years".

Project objectives (eg, what is the question you want to answer? What do you want to learn how to do? What do you want to find out?):

In this project, I intend on deepening and broadening my knowledge of quantum computers, a topic that has fascinated me as a potential new breakthrough technology, learning both how they work at technical level, and their key drawbacks. Quantum Computers are often touted in mainstream media as the future of computing, I aim to investigate whether they can deliver on this promise of a paradigm-shift in computational ability, which would affect everyone on earth in our increasingly online lives.

I will investigate three main possible scenarios:

1. Quantum computers prove an unviable technology.
2. Quantum computers prove useful only in an academic, on highly abstract problems, yielding limited impacts on mainstream computing.
3. Quantum computers have wide and deep effects that transform computing as we know it.

By doing this, I will better understand how viable they are as a technology in the next 10 years, and thus how pertinent they are to both future academic research and the everyday person.

Completing this Extended Project Qualification will allow me to further my knowledge on quantum physics, a topic which I'm passionately curious about, and intend to continue studying at university. Furthermore, by developing my dissertation, I will also attain several key academic skills:

- Sourcing reliable and relevant academic research
- Citing where relevant within my dissertation
- Collating information from a variety of sources into one cohesive dissertation
- Evaluating the reliability of various sources
- Using academic language to articulate my ideas.

Through the presentation section of the project, I will also improve my public speaking skills, developing my confidence in this area. The presentation will also require me to summarise and deliver what I've learnt to an audience, further developing my skills at explanation, in an accessible, yet technical manner.

If it is a group project, what will your responsibilities be?
This is not a group project.

Section Two: Reasons for choosing this project

Reasons for choosing the project (eg, links to other subjects you are studying, personal interest, future plans, knowledge/skills you want to improve, why the topic is important):

Completing the research for this Extended Project Qualification, will enable me to further develop my understanding of quantum physics, which is a field that I find fascinating, describing at a fundamental level how our world truly works, and the idea of this seemingly abstract branch of physics possibly having large impacts and applications that might potentially affect the future of computing, and thus humanity, excites me greatly.

Physics, and particularly quantum physics is fundamentally driven by curiosity and the thirst for knowledge, which appeals to an inquisitive person like me. Through studying this subject, I can quench some of that curiosity, whilst also attaining a deeper understanding of physics. This further enables me to understand in greater depth the practical applications, for, as opposed to pursuing knowledge for curiosity alone, this gives the pursuit of knowledge a purpose, not leaving it as some abstract endeavour.

Furthermore, completing an Extended Project Qualification, on physics will benefit me greatly, for physics is the subject which I aim to study at university, so, by undertaking a research project in physics, I will attain a great breadth of applicable skills and depth of knowledge. These skills and knowledge I gain I can apply during my university course, allowing me to understand the course content to a deeper level.

Having completed a Foundation Project Qualification and Higher Project Qualification, in previous years, I have a basic understanding of some of these skills, for instance: researching, citing, distilling research papers and collating information into something original and new. However, my current abilities with these skills are limited, and, by doing the Extended Project Qualification, I can improve, develop and deepen this skillset, which will serve me well in my future at university and beyond.

It is my opinion that this project is highly important and relevant to our increasingly computer-reliant society as a whole, given the possibility of this technology to impact and change the world, potentially a paradigm shift in computing, leading to leaps forwards in areas like artificial intelligence training, which itself will play a key part of our future.

Section Three: Activities and timescales

Activities to be carried out during the project (eg, research, development and analysis of ideas, writing, data collection, numerical analysis, rehearsal techniques, production meetings, production of final outcome, administration, evaluation, preparing for the presentation, etc):

I have chosen to break my timescales into various phases, in order to discretely segment the different stages of completing an EPQ. This will allow me to focus in on the most important parts of my EPQ in any given moment, for, as opposed to performing research during the planning phase, and planning during the essay phase, I can ensure that the tasks overlap minimally, ensuring I am able to focus all my attention on each phase. Furthermore, breaking down tasks into their appropriate phases, provides structure to completing seemingly impossibly large tasks, ensuring such large tasks are broken down and decomposed into smaller, more achievable activities.

Planning phase:

- Deciding on provisional essay title

How long this will take:	
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The planning phase should take 1- 2 weeks

- Complete Project Proposal Form
- Using resources like Google scholar and JSTOR to investigate whether the area of study is too niche or too broad given the availability of academic research papers on the topic.

Research phase:

- Gathering research papers
- Distilling research into notes
- Condensing notes
- Analysing research with an open mind
- Analysing sources based on bias, age, and reliability.

Essay writing phase:

- Decide on essay outline.
- Forming an informed opinion on my topic (quantum computing's potential impact)
- Collating notes into an evidence-based argument supporting my point of view.
- Considering the merits of other perspectives on the matter, and what evidence there is to oppose my argument.
- Possibly researching deeper into areas that need more evidence or backup.
- Transforming my ideas and notes into a coherent, informed, academically written argument.
- Citing my essay properly to provide evidence behind any stated facts.

Review phase:

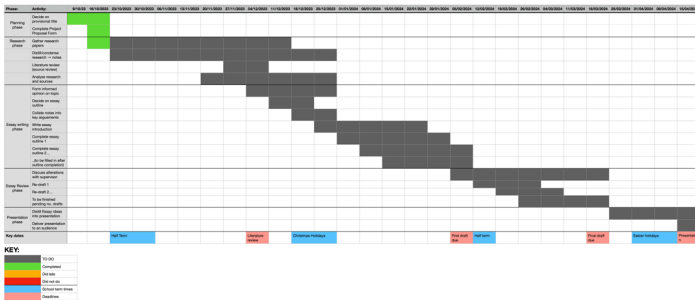
- Discuss with my supervisor any potential improvements that could benefit my essay.
- Re-draft my essay with the required alterations

Presentation phase:

- Distil my ideas and arguments from essay form into informative, evidence-based persuasive presentation.
- Deliver this presentation to a group of peers and teachers.

In total, this project should take me up to 29 weeks to complete, which I have outlined in a Gantt chart (below) to ensure that I'm held accountable to my schedule and that I do not fall behind on any one area or spend too many resources and time on one specific phase.

EPQ Gantt chart



maximum (overlapping with the research phase)

The research phase should take 10 weeks maximum, heavily overlapping with the essay writing phase.

The essay writing phase should take 10 weeks, overlapping with the research phase.

The review phase should last for 6 weeks, overlapping with the essay writing phase.

The presentation phase should take 3 weeks maximum, overlapping with the completion of the final draft in the review phase.

Milestone one: Project Proposal Form
Target date (set by tutor-assessor): 20/10/23.

Milestone two: Literature Review
Target date (set by tutor-assessor): 08/12/23.

Milestone three: First Draft
Target date (set by tutor-assessor): 09/02/24.

Milestone four: Final Draft
Target date (set by tutor-assessor): 22/03/24.

Milestone five: Presentation
Target date (set by tutor-assessor): 15/05/24

Section Four: Resources

What resources will you need for your research, write up and presentation (eg, libraries, books, journals, equipment, rehearsal space, technology and equipment, venue, physical resources, finance):

I will use a wide variety of resources for my research, though I aim to mainly focus my attention on academic journals and well-regarded but more popular publications (like the New Scientist). The journals will mainly be sourced from resources such as:

- Google Scholar
- Google books
- Microsoft Academic
- WorldWideScience
- Springer Link
- DOAJ and DOAB (Directory of Open Access Journals/Books)
- JSTOR
- Open Access Button
- Any other free to access publications.

I hope to use as many primary sources as possible, though will not abstain from utilising well cited, up to date secondary sources.

For each source, I will also analyse the sources reliability based on factors including:

- Writer/publication bias - does the author/publication hold any vested interest in quantum computing?
- Date - quantum computing is a highly dynamic fast-moving field, so any information that's written too long ago may possibly be outdated or irrelevant to the current state of quantum computing
- Number of citations and Peer review, which allows me to check both the trustworthiness and relevancy of a paper, as well as the quality of the information within.
- The qualifications of the author - is the author qualified to deliver any assertions they may provide

I also aim to make use of our local library in order to access physical books which can not only provide insight into quantum computing's potential impacts, but also yield explanation of the highly complex physics behind quantum computers.

What your areas of research will cover?

My research will cover the foundations of how quantum computers work, in order to fully understand the reasoning behind their potential to impact computing, or otherwise within the next decade.

I shall then move on to researching the actual viability and thus impact that quantum computers could have, basing this research on the knowledge I gained during the first stage.

Within the discussion of impact and viability, I intend on researching three main areas of possible areas:

- Their utility as a tool for academics, impacting only how we do science, but capable only of solving abstract problems.
- The possibility quantum computers go on to completely revolutionise modern computing, replacing conventional binary based computer systems with more power and efficiency, and thus proving to be highly adaptive to a broad range of tasks.
- The technologies that quantum computer might *enable*, possibly indirectly leading to the creation of many revolutionary technologies in their own right - just as the conventional “binary” computer enabled an internet and so much more, what could a quantum computer enable?

Comments and agreement from tutor-assessor

Is the learner taking this project as part of the Diploma?

Yes/No

If yes, which Diploma are they taking? _____

Comments (optional):

Is project derived from work which has been/will be submitted for another qualification?

Yes/No

Which qualification (title and unit)? _____

Comments (optional):

I confirm that the project is not work which has been or will be submitted for another qualification and is appropriate.

Agreed:

(name)

(date)

Comments and agreement from project proposal checker

Comments (optional):

I confirm that the project is appropriate.

Agreed:

(name)

(date)

Project Activity Log

Learner Name Daniel Fussey Learner number 1593
Centre Name Charters School Centre Number 51529
Unit Name P301 Unit number 06/10/2023

Teacher Assessor _____

Proposed project title To what extent will quantum computers have an impact on computing in the next 10 years

This form should be used to record the process of your project and be submitted as evidence with the final piece of work.

You may want to discuss:

- what you have done (eg, from one week to the next)
- if you are working in a group, what discussions you have had
- any changes that you have or will need to make to your plans
- what resources you have found or hope to find
- what problems you are encountering and how you are solving them
- what you are going to do next

Date	Comments
Week 1 02/10/2023	During this week we held our session on the EPQ and Project Proposal Form, and were given detailed advice on what to include in each section. Furthermore, we laid down internal deadlines to meet for each stage of our project (literature review, main essay, and presentation). I also began searching for a topic by researching possible regions of interest, mainly within physics - the subject I hold the most passion for. I intend to use Google Scholar in order to research the possible depth and breadth available for any candidate topics. The current two main topics I am interested in are quantum computing, and nuclear fusion.
Week 2 09/10/2023	After a period of initial research into both topics, I have decided to pursue quantum computing. This is both due to its accessibility to someone of A-level physics and maths ability, as well as its relevance and media prominence. I then decided upon a preliminary title ("To what extent will quantum computers have an impact on computing in the next 10 years") which I believe has both a possibility for great depth of research whilst restricting scope to a realistic level. I have completed section one of the project proposal form, using the advice and tips given in our last EPQ meeting.
Week 3 16/10/2023	During week three of my EPQ, we held a meeting on research techniques. Following this meeting, I intend to utilise other sources beyond google scholar, including JSTOR, Microsoft Academic and more, which will hopefully enable me to cast a wider net when sourcing research papers. We were also taught skills in critiquing sources, checking for their relevance, and for bias, which will both help the research itself, as well as giving greater detail into the literature review. I have, on my Project Proposal Form, completed to section two (about my motivations for choosing the project), and made a start with the third section (timescales).
Week 4 23/10/2023	I have decided to augment my proposed project title from "To what extent will quantum computers have an impact on computing in the next 10 years" to "To what extent will quantum computation develop into a transformative technology within the next decade?" I believe that this has greater scope for discussion of quantum technologies, shifting the focus from the quantum computers themselves towards the technologies they enable. However, the majority of the work completed in this week was gathering research papers, primarily through google scholar. I have managed to source 20 different sources on quantum computing that are relevant to my project.

Week 5 30/10/2023	This week I completed and submitted sections 3 and 4 of my project activity log, creating a Gantt chart in section 3 (timescales) in order to plan and manage my time going forwards. I then combed through the sources I found last week, by scanning their abstracts, and have decided to begin research by analysing the source "Introduction to quantum computing for non-physicists". I believe this source will allow me to understand the fundamentals of quantum computing, which will be greatly beneficial going forwards as I read more complex sources. I found the start of the paper incredibly accessible, with it describing the basics of quantum measurement and superpositions through the use of photon polarisation experiments.
Week 6 06/11/2023	Week six of my project was used for deeper analysis of "Introduction to quantum computing for non-physicists" which I have found very useful. Whilst it is not a modern paper, having been written in 2000, it has 178 citations, some very recent, which suggests the underlying mechanisms by which quantum computers are built, then largely theoretical, have changed little. The section I analysed concerned the basic Ket notation that quantum computing papers use to describe states, as well as a primer to the underlying maths of single qubits (quantum bits).
Week 7 13/11/2023	We held an EPQ session on the dangers of plagiarism and proper referencing using the Harvard referencing scheme. This will be useful when I start writing my essay as it will allow me properly credit the work I cite. The majority of week seven was used reading the paper "introduction to quantum computing for non-physicists", where I learnt about the maths behind multiple qubits, and how this leads to quantum computational speed-ups. Whilst this paper is taking a long time to properly analyse, I believe, as it will provide a foundational understanding with which I can approach more complex papers, it is worth the time investment in deeply understanding.
Week 8 20/11/2023	This week I completed the relevant sections on "Introduction to quantum computing for non-physicists", which delved into quantum logic gates and, crucially the new concept of quantum parallelism. This is of great interest to me as it appears to be the root source of quantum computational advantage. I have therefore set aside quantum parallelism as a topic to research into greater depth at a later date. I have also made a start on the paper "Present landscape of quantum computing", which I hope to use to both fact-check the previous paper for inaccuracies (Intr. to QC for non-physicists being from 2000), and to gain insight into the current state of the field.
Week 9 27/11/23	We held our EPQ session on the literature review. We were instructed not to review all our sources, but only the most important 4 or 5 that will contribute most to our argument. We were also given a format to follow. As my research papers are all from independent journals, problems of bias are unlikely to be an issue, however, due to the scientific nature of my project, methodology and reliability are likely to come into play.
Week 10 04/12/23	In this week I completed research on the paper "Present landscape of quantum computing". This paper provided really useful summaries on the current cutting edge, including various different methods of qubit production being pursued, such as superconducting qubits (used by Google and IBM) and trapped ions (used by start-up IonQ). This dispelled the notion I initially held that there was only one viable method to produce quantum computers, however, as my essay is concerned with quantum computing as a holistic whole, I believe the merits of different methods are beyond my scope.
Week 11 11/12/23	During week eleven, we held an EPQ meeting on further research methods, which included tips for note-taking and summarising notes that we already had. After I finished reading "Present landscape of quantum computing" I abstracted the notes I currently had from both sources into a set of general notes from all sources. This will give me one place to look at each sub-topic, rather than having to jump around different sources - bringing greater clarity to my thought process.
Week 12 18/11/23	I continued my literature review, discussing "An introduction to quantum computing for non-physicists". Whilst this paper is very old, my conclusion was that the physics included was still valid and thus relevant. However, whilst this paper gave me a deep understanding of quantum computers, and was thus valuable, I don't believe that it contains much information relevant towards my project. I further fully researched the paper "Quantum computing - A taxonomy of systematic review and future directions".
Week 13 25/11/23	I have decided my current rate of research is too slow, thus, in order to accelerate research, I will be dedicating more time towards individual applications, rather than the more broad but shallow papers that I have used thus far. To this end, I have studied quantum AI this week, using the papers "Quantum computing and AI: Impacts and possibilities", and "Quantum computation, quantum theory and AI". I also completed my literature review this week and sent it to my supervisor to review.

Week 14 01/01/2024	This week I made a start on my essay, which I have decided to divide into two main rough sections: 1. where quantum benefits arise from 2. what they can be used for I have also met with my supervisor to discuss my literature review, where the main comment was on the word count. My supervisor is worried that I won't have enough words left over to fill out my essay, keeping it around 6000 words, as the literature review is currently in excess of 2000 words. We discussed potentially removing one of the sources if needed.
Week 15 08/01/2024	Week 15 was used for writing the fundamental physics sections of my essay, from qubits and superpositions to measurement and entanglement. I found explaining these sections difficult, as figuring out the appropriate level of assumed knowledge I intended my reader to possess, and then writing to that standard presented a challenge I had never come across before. I also used this week to analyse a variety of new applications, notably enhanced drug development, which surprised me with its potentially massive impact, and the suitability of quantum computers for the role.
Week 16 15/01/2024	This week I completed my essay section on the building blocks of quantum computers, by writing about quantum scaling, quantum parallelism (which I took the time to explore in greater detail), and quantum gates/circuits. This section was most striking for its many resemblances to classical design, which facilitated straightforward comparison between the two.
Week 17 22/01/24	I have realised that I require a greater number of applications, and so decided to spend week 17 researching a wide range of application, using the papers "Commercial applications of quantum computing" and "Forthcoming applications of quantum computing". These gave me enough breadth to hopefully continue with analysing applications next week
Week 18 29/01/24	This week I had to write my section on applications twice. The first attempt was corrupted by Microsoft Word, and, as I had no more recent backups, I was forced to re-write the entire 2000-word section. Fortunately, in doing so, I was able to better categorise and organise the section, developing a system of 3 main application types: Type 1: Direct benefits. Type 2: Quantum enabled classical technologies. Type 3: Quantum enabled novel technologies. I believe this improved structure will greatly benefit my essay.
Week 19 05/02/24	Our internal deadlines listed week 19 as the deadline for submitting my first draft. Due to the corruption of my file, I have been unable to complete and abstract and fully cite my essay in time, however, I have managed to write a balanced conclusion and introduction, which, along with the main essay body, have been submitted to my supervisor for review.
Week 20 12/02/24	On week 20, I showed my essay to my parents, and implemented suggestions they inputted about the structuring and wording of the first section.
Week 21 19/02/24	This week we held out EPQ session on presentations. We were told what we ought to include within our presentations and given tips on how to act. The structure I intend to follow is a 1-minute introduction, followed by a 7-minute discussion of quantum computers, and a 1.5-minute conclusion. I have also made a start on creating my presentation, with an outline of how I intend to use the time I have to best convey such a complex topic.
Week 23 04/03/23	On week 23 I met my supervisor to discuss my first draft. They were highly supportive of the content itself, but also concerned about the word count, which currently sits 2000 words above the limit. They suggested moving some of the explanation section into an appendix, which does not contribute towards word count, and suggestion I intend on implementing in the near future.
Week 24 11/03/23	Week 24 was dedicated towards implementing the suggestions of my supervisor. Having checked the targets, we realised that the 6000-word target was not a limit, and as such my longer word-count was less of an issue. I further did a second pass for citations, filling in gaps that I had missed. If I were to repeat my project, I would ensure that citations were added as I wrote it, for catching up on sources after the fact was time consuming.
Week 25 18/03/23	This week was the week of final submission. I spent time ensuring that my PPF (Project Proposal Form) and PAL (Project Activity Log) was in order and copied them over into the final master document. Once submitted I will make a start on my presentation.